

PATIENT MULTI-OMICS GENOMIC HEALTH SUMMARY

Patient Submitter ID: TCGA-BC-A108 | Tumor: Liver Hepatocellular Carcinoma (LIHC) | Stage: Stage I

TCGA Project:	TCGA-LIHC	cBioPortal Study:	lihc_tcga_pan_can_atlas_2018
Gender / Age:	Female / 58 yrs	Vital Status / OS:	Alive (44.1 mo)
Hybrid ML Risk:	74.1% (HIGH RISK)	VQC / Classical:	71.0% / 76.2%

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

Gene	Locus (GRCh38)	Transcript	Protein Change	ClinVar Status	COSMIC ID	gnomAD Pop AF	VAF %
TP53	chr17:17p13.1	ENST00000269305	p.R249S Missense Mutation	Pathogenic VCV000012384	COSV52668586 FATHMM: 0.99 Pathogenic	0.000001	37.8%

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ_VQC / R_Y / CNOT)

Gene Symbol	Canonical Ensembl ID	VAF Hotspot Frequency	Variational Angle θ (Radians)	Qubit Hilbert Space Projection
TP53	ENSG00000141510	37.8%	θ = 1.9420 rad (111.3°)	$ \psi\rangle = \cos(0.971) 0\rangle + \sin(0.971) 1\rangle$

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:
Surgical segmentectomy resection followed by atezolizumab plus bevacizumab immunotherapy surveillance and CT dynamic liver monitoring.

Genomic Biomarker Summary: Identified somatic hotspot mutations in TP53 (p.R249S). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the BlazeFinix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.